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The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

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HOSUR-635109

Induction Program(24th to 26th Sep.2025)
Day 1-24.09.2025
Overview of research
Prod.G.L.Samuel
Dept of Mech Engg-IIT Madras

What is research?.

Research may be described as a structured and organized process of inquiry aimed at discovering new knowledge and establishing facts and concepts systematically.

Research:

- A structured and methodical study conducted to verify and establish factual information.
- An intellectual process aimed at arriving at new conclusions or finding effective solutions to problems.

Why do you want to do research?

- Quest for Knowledge
- Must for a career
- The process is fun and exciting.
- Seeking recognition in a peer community.
- Don't know, but I've drifted into it.
- Nothing else to do.

Types of research:

- Fundamental Research
- Applied Research
- Descriptive Research
- Analytical Research
- Experimental Research

Basic Research (also referred to as pure or fundamental research) focuses on enhancing existing scientific knowledge for its own value, rather than for any immediate practical use. It is motivated by intellectual curiosity and the aim to develop, refine, or advance theories.

Example: Investigating the fundamental components of atoms or how different types of cells communicate.

Applied Research uses existing scientific theories and knowledge to solve specific, practical, real-world problems. Its goal is to provide immediate and workable solutions to concrete issues.

Example: Researching ways to improve the energy efficiency of homes or developing a new vaccine for a specific disease.

By Methodology/Objective

Descriptive Research aims to clearly and precisely describe the features and attributes of a specific individual, group, or situation. It answers the "what is" question, but generally not the "how" or "why". This method often uses surveys and observational methods.

***Example*:** A study describing the frequency of grocery shopping preferences among a specific demographic.

Analytical Research involves examining existing data, facts, or information in order to interpret them and carry out a critical assessment.It involves in-depth study and critical thinking to explain complex phenomena and understand underlying causes or relationships (answering "why" and "how").

***Example*:** Analyzing existing economic data to determine the factors that caused a trade deficit to move in a certain direction.

Experimental Research is a systematic and scientific approach used to establish a cause-and-effect relationship between variables by manipulating one or more independent variables in a controlled environment.

Example: In a clinical study, one set of participants is given a newly developed drug (experimental group), while another set receives a placebo (control group) to evaluate the effectiveness of the medication.

Case Study 1: Deep Learning in Precision Agriculture (Crop Disease Detection)

Problem:
Farmers struggle to detect crop diseases early, leading to lower yield and increased pesticide use.

Deep Learning Approach:
Researchers collected thousands of leaf images from crops (e.g., tomato, potato, maize).
They trained **CNN (Convolutional Neural Network)** models such as **ResNet, VGG16, and MobileNet** to classify diseases from leaf images.

Key Steps:

- Image preprocessing (noise removal, augmentation)
- Model training on labeled disease images
- Evaluation using accuracy, precision, recall

Outcome:

- Models achieved **90–99% accuracy** in identifying diseases like blight, leaf spot, and rust.

- Farmers could use mobile apps to scan leaves and get instant disease diagnosis.
- This improved early detection and reduced pesticide overuse.

Impact:
Low-cost and scalable solution for improving crop health and yield.

Key Differences between Research Method and Research Methodology

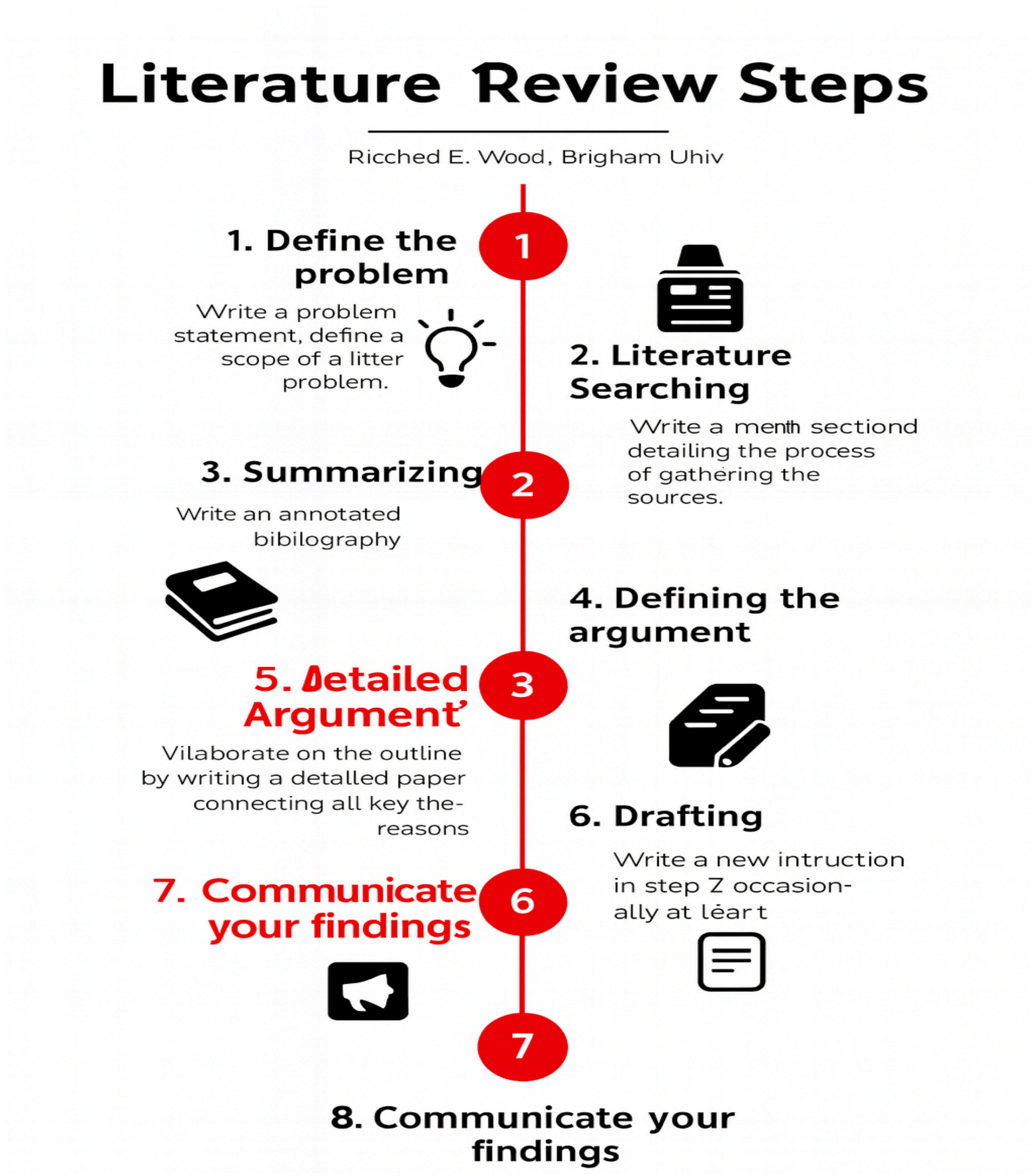
1. Research methods concentrate on the specific tools and techniques used for collecting data, whereas research methodology refers to the broader strategy and structural plan of a study.
2. A research method consists of concrete procedures and actions taken to gather information, while research methodology helps the researcher decide which methods are most suitable for the study.
3. Research methods define the kind of data to be collected and how it will be analyzed, whereas research methodology offers a structured and organized approach for carrying out the research process.
4. Common research methods include surveys, interviews, experiments, and observations, while research methodology includes qualitative, quantitative, and mixed-method approaches.
5. A research method explains how data are collected and analyzed, whereas research methodology addresses the theoretical, philosophical, and conceptual foundations of the research.
6. Research methods are practical and action-oriented, focusing on implementation, while research methodology is more conceptual, emphasizing research theory and rationale.
7. Research methods form a component of research methodology, which covers the entire research framework from planning to execution.
8. A research method is usually tailored to a specific study, whereas research methodology can be applied to various research projects across disciplines.
9. Research methods influence the reliability and accuracy of collected data, while research methodology ensures the overall credibility, rigor, and coherence of the research.
10. Research methods deal with the specific tools and techniques used for data collection, whereas research methodology emphasizes the overall approach and structural framework guiding the research process.

Research Process:

What is literature review ?

A literature review is a survey and critical synthesis of existing scholarly work (books, articles, etc.) on a specific topic, providing an overview of current knowledge, identifying key researchers/theories, and highlighting gaps your own research can fill, often forming part of larger paper like a thesis or acting as a standalone piece.

Literature Review Steps:



Induction Program(24th to 26th Sep.2025)

Day 2-25.09.2025
Research to Startups
Dr.G.S.Mahalakshmi
Director – CED AU

🎬 **Bridging Research and Entrepreneurship**: The session emphasizes converting research outcomes (ideas, findings, innovations) into real-world products/services through startups. It encourages researchers to think beyond academic publications — to consider entrepreneurial and societal impact.

🎬 **Opportunities in Innovation & Commercialization**: Dr. Mahalakshmi highlights that academic research can lead to patenting, product development, or startup creation — offering a pathway for researchers to turn their work into market-ready solutions.

🎬 **Support Ecosystem at Anna University / CFR**: The induction programme (and the Centre for Research / Entrepreneurship Development at Anna University) offers guidance, resources, and frameworks to help budding researchers-entrepreneurs — implying that research does not end at thesis, but can extend to innovation/startup.

🎬 **Mindset Shift for Researchers**: The talk likely encourages a mindset shift: from "research for knowledge" to "research for impact" identifying problems, innovating solutions, and considering social or commercial relevance.

Induction Program(24th to 26th Sep.2025)
Day 2-25.09.2025
Intellectual Property Rights
Dr.M.Kanthababu-Director-CFR,AU

1. Fundamentals of Intellectual Property (IP)

- **Invention vs. Innovation:** An **Invention** is finding something for the first time in the world, while **Innovation** is improving upon an existing system.
- **What is a Patent?** A patent is a document issued by the government for any new **innovation** or **invention**. It grants exclusive rights to the inventor and is typically valid for 20 years, after which the idea enters the public domain.
- **The Concept of "Rights":** Registering an idea at the Intellectual Property Office grants the inventor ownership rights, similar to registering a car at the RTO or a property at the Sub-Registrar's office. This allows the inventor to stop others from using the idea.

2. The Three Criteria for Patentability (Innovation)

For any research work, including a PhD thesis or a product, to be considered truly innovative and patentable, it must satisfy three criteria:

1. **Novelty:** The idea must be new to the world. It must not have been published previously in a journal, conference paper, or disclosed on social media like Facebook or WhatsApp. If you disclose your idea, you lose novelty.
2. **Inventive Step (Technical Advancement):** The innovation must involve a non-obvious technical advancement or improvement over existing products or processes. For example, moving from a fixed mic to a wireless mic is an inventive step.
3. **Application:** The patented idea must have a demonstrable industrial or real-world application.

Example of the Stumps: The lecture uses the example of the LED stumps used in cricket (Zings) to illustrate these points: the novelty was having light glow when the bails were separated; the inventive step was the electronic circuit, battery, and sensors integrated into the bails and stumps; and the application was for cricket matches.

3. Advice for Researchers

- **Patent First, Publish Later:** Researchers are strongly advised that as soon as they develop a prototype or get an innovative idea, they should file a patent first before publishing a paper. This protects their idea and allows them to earn money through royalties or selling the technology.
- **Search Patent Literature:** The speaker highlights that 90% of good literature and commercially valuable information is found in patent offices, not in traditional search engines like Google Scholar. Researchers must search patent databases to ensure the novelty of their work
- **Product Focus:** Patents are typically given for products or processes, not methods of use (e.g., the "helicopter shot" in cricket is not patentable).

4. Other Types of IPR

The lecture briefly describes other forms of intellectual property:

- **Industrial Design:** Protection for how a product looks (its aesthetic features), such as the unique shape of a Coca-Cola bottle.
- **Trademark:** Protection for a brand name, logo, or symbol (e.g., McDonald's 'M' or the Starbucks logo) to prevent others from using the same identity for similar goods or services.
- **Trade Secret:** Information that gives a business a competitive advantage and is kept confidential rather than being registered (e.g., the formula for a coffee blend).
- **Copyright:** Protection for literary, dramatic, musical, or artistic works (e.g., a song, a book, or even a thesis).
- **Geographical Indication (GI):** Identifies a product that originates from a specific geographical region and possesses a reputation or characteristics attributable to that region (e.g., a specific type of sweet or textile).

5. Startup vs. Entrepreneurship

A clear distinction is made between an entrepreneur and a startup founder:

- An Entrepreneur typically manufactures and sells an existing product (e.g., a standard fan), competing primarily on price.
- A Startup Company is founded on a product that has innovation (e.g., an IoT-based or voice-activated fan). The innovation is protected by a patent, which provides a 20-year monopoly, preventing competitors from simply copying the product and undercutting the price .

Induction Program(24th to 26th Sep.2025)
Day 2-25.09.2025
Fellowship Opportunities
Dr.C.Arun Prakash
Deputy Director-CFR,AU

What is fellowship?.

A fellowship is a temporary, funded program for advanced study, research, or training, often for post-graduates, offering financial support and practical experience in fields like medicine, academia, or public policy to develop specialized skills beyond basic qualifications, essentially a step up from internships but focused on deep expertise.

Anna Research Fellowship:

- 3 years (or till submission of thesis) with effect from the date of joining the Ph.D. course in the department.
- No other part-time job/any other fellowship/project assistantship.
- Signing in attendance register on all working days.
- Assist supervisor in the academic work(practical sessions, tutorial periods, and other related academic activities) not exceeding 8 hours per week.
- Progress report for every 6 months.
- Stipend – Rs. 31,000 + 24% HRA (= Rs. 38,440) per month
- A fellow may not discontinue his or her fellowship unless prior approval is obtained from the Director of the Centre for Research. The fellowship shall officially conclude on the date the resignation takes effect.

- A duly maintained attendance record, signed by the research supervisor and endorsed by the Head of the Department or the Director of the Centre, must be submitted together with the six-month progress report.

- Prior approval from the supervisor and the HoD or Director of the Centre is required to attend national conferences, seminars, fieldwork, or similar academic activities, which will be considered as "On Duty."

- The approved permission documents must be submitted to the Centre for Research when applying for the fellowship stipend for the respective month.

Leave rules

- An Anna Research Fellow is permitted to take leave for up to 20 days in a calendar year, excluding Saturdays (where applicable), Sundays, and declared public holidays.

- Anna Research Fellows are not entitled to medical leave or any additional categories of leave beyond the permitted limit.

- Women Fellows who have fewer than two living children are entitled to receive their full stipend during maternity-related absence, for a maximum period of 90 days, once during the fellowship tenure.

ARF Review

- An expert committee appointed by the Vice-Chancellor will conduct an annual technical assessment of the ARF scholar's research progress and academic achievements.

- Renewal of the fellowship for the third year will depend on the expert committee's recommendation, which will be based on the scholar's research performance.

- The ARF scholar must present his or her research findings at a National or International Conference during the second year of the fellowship.

- The scholar is required to publish one research paper related to their work in a journal approved by the Centre for Research.

ADF

- Engineering and Technology
- Management
- Design
- Planning
- Applied Science
- Hotel Management and Catering Technology

Registration

- Category (A): Applicants can apply directly to the scheme through the AICTE ADF portal by submitting a No Objection Certificate (NOC) from the concerned University, as per the scheme's guidelines.

- Category (B): Universities may register under the scheme and admit candidates to their Ph.D. programs in accordance with their institutional admission process and the minimum eligibility criteria prescribed by AICTE.

Age

- The applicant must not have completed 30 years of age at the time of admission.

- An age relaxation of up to 5 years is permitted for candidates belonging to SC, ST, Women, and Persons with Disabilities, while a relaxation of 3 years is applicable for candidates from the OBC category.

- If a candidate qualifies under more than one category, the total age relaxation allowed shall not exceed 5 years.

Induction Program(24th to 26th Sep.2025)
Day 3-26.09.2025
eTools for Researchers
Prof.S.Indran
Dept of Mech Engg-Alliance University,Bengaluru

Key eTools and Tricks for Researchers

1. Selecting the Right Journal

- **Post-Manuscript Selection:** After writing an article, researchers can use tools like the **Elsevier Journal Finder** (and similar tools from IEEE, Sage, etc.) by entering the title and abstract to get a list of best-match journals.
- **Pre-Manuscript Selection:** It is highly recommended to choose the right journal *before* starting to write the article.
- This involves:
  - Finding a "guruji" (expert) in your field who frequently publishes articles.
  - Tracking their publications to see which journals they use consistently.
  - Focusing on a few high-quality journals with a matching aim and scope, rather than trying many different ones
- **Prioritizing Speed:** For Ph.D. scholars, Professor Indran suggests prioritizing the journal with the fastest "Time to First Decision" to get a quick acceptance or rejection, which saves time.

2. Fast and Efficient Manuscript Typing (Voice Typing)

- The speaker encourages researchers to save time by writing the first draft without using their fingers.
- **Google Docs Voice Typing:** Use the "Tools" > "Voice typing" option in Google Docs to dictate content.
- **Microsoft Office 365 Dictate:** Use the "Dictate" feature in Microsoft Word 365.
- Tip: Speaking one's observations in the lab and using a noise-cancellation headset to dictate the content directly helps create a "free flow writing" style that inherently helps avoid plagiarism Networking and Collaboration

- **Google Scholar Alert:** Follow leading researchers in your area on Google Scholar and turn on the "Follow" alerts for new articles and citations. This helps track which journals these top researchers are publishing in.
- **Research Gate:** Create a ResearchGate account using your institutional email ID
  - **Follow Researchers:** Follow key individuals and check the "Stories" feed for their new publications.
  - **Connect:** Use the platform to message world-class researchers (especially those who read or cite your work) to build a network .This networking is crucial for post-doc and international opportunities.

3. High-Quality Pictures for Publication

- Reviewers check the quality and structure of pictures and figures immediately
- IrfanView Software**:** The speakers recommend using the lightweight (4 MB) IrfanView software to enhance and manage images.
- **Workflow:**
    1. Prepare all figures, charts, and diagrams in PowerPoint.
    2. Save the slides as an Enhanced Windows Metafile (EMF).
    3. Open the EMF file in IrfanView.
    4. Use the Batch Conversion feature to convert the image to the TIFF (Tagged Image File Format), which is preferred by high-quality journals.